

Examining the k th component before and after the transformation leads to the requirement

$$\frac{g_f(j, k)}{g_f(i_0, k)} = G_f(j), \quad (6)$$

where $G_f(j)$ is independent of the column index k . This condition is equivalent to the functional separability of g_f [20]. If not separable, the transformation generally modifies the number of linear dependencies among rows or columns, altering the rank. Conversely, separability ensures a one-to-one correspondence between null vectors of A and B , guaranteeing identical rank and nullity. A complete derivation is provided in Appendix A.

C. Transformation of Zero-Mode Profiles

We now determine how the zero modes themselves transform under a separable scaling. This result is crucial for understanding how localization profiles can be engineered without affecting the existence of massless fermions.

Zero-Mode Transformation. Let $\{v^1, \dots, v^n\}$ be a basis for the null space of A . Under a separable transformation $g_f(i, j) = g_f^{(L)}(i) g_f^{(R)}(j)$, the corresponding null space of B is spanned by vectors v^i whose components are

$$(v^i)_j = (v^i)_j g_f^{(R)}(j). \quad (7)$$

This follows from substituting $a_{lj} = b_{lj} g_f^{(L)}(l) g_f^{(R)}(j)$ into the condition $Av^i = 0$, yielding

$$\sum_j b_{lj} (v_j^i g_f^{(R)}(j)) = 0, \quad (8)$$

which shows that $v_j^i = v_j^i g_f^{(R)}(j)$ spans the null space of B . The column-dependent factor $g_f^{(R)}(j)$ therefore directly modifies the spatial profile of the right-chiral zero mode, while the row-dependent factor $g_f^{(L)}(i)$ analogously controls the left-chiral profile (as relevant for Dirac fermions). This independent row/column action enables controlled reshaping of the two chiral sectors while preserving the zero-mode count.

Eigenvalue Preservation. For completeness, we note that if A is diagonalizable with eigenvalues $\{\mu_i\}$, then B shares the same eigenvalues provided the separable function satisfies

$$g_f^{(L)}(k) g_f^{(R)}(k) = 1 \quad \forall k. \quad (9)$$

This additional constraint ensures that diagonal entries transform trivially. The proof is given in Appendix A.2.2. Although eigenvalue preservation is not required for our later applications, we include it as it clarifies the broader mathematical structure of the transformation.

Having established the rank-preserving nature of separable transformations and their action on zero-mode profiles, we now turn to explicit applications within the context of theory-space models.

III. APPLICATIONS TO THEORY SPACE MODELS

In this section we illustrate how rank-preserving separable transformations may be used as a practical tool for engineering fermion mass structures in theory-space models. Since the transformations preserve the number of zero modes while reshaping their localization profiles, they enable controlled manipulation of both the chiral structure and the massive spectrum. We present several representative applications. Section III A shows that the standard clockwork construction arises as a special case of an appropriate separable function, while more general choices lead to qualitatively different localization patterns. Section III B demonstrates how the independent row/column action of the transformation allows one to separate the left- and right-chiral zero-mode profiles. In Section III C we examine the impact on Kaluza–Klein spectra in dimensional deconstruction. Finally, in Section III D we apply the framework to neutrino mass generation as a concrete phenomenological example.

A. Generalizing the Clockwork Mechanism

The clockwork mechanism provides a particularly elegant realization of large hierarchies from $\mathcal{O}(1)$ parameters by generating exponentially suppressed effective couplings through zero-mode localization in a discretized extra dimension or theory space [1, 3]. In its standard form, a nearest-neighbor structure produces a single massless mode whose wavefunction is exponentially localized toward one end of the “gear chain,” thereby inducing hierarchical interactions with fields located on specific sites [21]. This structure is usually introduced at the level of the Lagrangian via site-dependent couplings or background parameters.

Our framework reproduces the clockwork construction as a special case of a rank-preserving separable transformation and simultaneously provides a method for exploring a broader space of localization profiles. To illustrate this, consider the generalized clockwork Hamiltonian with site-dependent masses m_i and nearest-neighbor couplings q_i [22],

$$\mathcal{H}_{i,j}^{\text{cw}} = m_i \delta_{i,j} + m_i q_i \delta_{i+1,j}. \quad (10)$$

This follows from the Lagrangian

$$\mathcal{L}_{\text{NP}} = \mathcal{L}_{\text{kin}} - \sum_{i=1}^n m_i \bar{L}_i R_i - \sum_{i=1}^n m_i q_i \bar{L}_i R_{i+1} + \text{h.c.} \quad (11)$$